The Nutrient Brief

EDITION 01 · MICRO

MINERAL OF THE WEEK

Magnesium Glycinate — your nervous system's *volume knob*.

Roughly half of adults don't get enough magnesium from diet. Here's what it actually does in your body, what the evidence really supports, and who tends to run low.

THE GAP

~50% of adults fall short on dietary magnesium. It's a cofactor in more than **300** enzymatic reactions — from muscle contraction to nerve signaling to how your cells generate energy.

HOW IT WORKS

Brakes and accelerators, in the brain.

ACCELERATOR
NMDA receptors — magnesium keeps them regulated.

BRAKE
GABA activity — magnesium helps activate it.

Low magnesium means weaker brakes *and* louder accelerator at the same time. That's why the classic symptoms cluster around jumpiness — poor sleep, anxiety, muscle twitches, migraines.

WHAT THE EVIDENCE SAYS

Strong where it counts, weak where it's hyped.

- Sleep quality, older adults
- Blood pressure (~2–3 mmHg)
- Migraine prevention
- Anxiety (small trials)
- Athletic performance, "energy"

MOST LIKELY TO BE RUNNING LOW

Who tends to benefit most.

The people whose baseline is most likely below the sufficiency line — so correction tends to show the biggest benefit.

Adults 60+
Absorption efficiency declines with age; intake often drops too.

Long-term PPI users
Omeprazole, pantoprazole & others deplete magnesium over months.

Type 2 diabetes & prediabetes
Higher urinary magnesium loss; gap compounds with insulin resistance.

Refined-cereal diets
Common in urban India; whole grains & leafy greens are the missing source.

Resistance trainers & endurance athletes
Sweat loss + heavy use in muscle contraction and recovery; training volume raises need.

Chronic stress / poor sleepers
Sustained stress increases urinary excretion; low mag in turn worsens sleep.

FORMS & AMOUNTS

Not all forms absorb the same.

Glycinate	~25–35%	High absorption, gentle on the gut. Commonly chosen for sleep & anxiety use cases.
Citrate	~25–30%	Good absorption, mildly laxative at higher doses (sometimes the point, sometimes unwanted).
L-Threonate	~20–25%	Crosses the blood-brain barrier more readily. Cognitive/sleep focus. More expensive.
Oxide	~4%	Cheap & common in mass-market supplements; mostly passes through.

Dietary reference: ~310–420 mg of elemental magnesium per day for adults (varies by sex & age; ICMR / NIH ranges).
Trial ranges: studies on sleep, blood pressure, and migraine typically used **200–400 mg elemental** per day of a well-absorbed form. Splitting doses (morning + evening) improves tolerance.

BEFORE YOU CONSIDER IT

Who should think twice.

Kidney disease
Impaired kidneys can't clear excess — high intake becomes genuinely dangerous.

Certain antibiotics
Fluoroquinolones & tetracyclines — space doses apart to avoid absorption interference.

EXPERT CONTEXT

An Indian clinician's view.

> *"Urban Indians on refined-cereal diets often fall short on magnesium — and for someone with prediabetes or type 2 diabetes, that gap matters more, because higher urinary magnesium loss compounds with insulin resistance."*
>
> — Paraphrased from the published research of **Dr. V. Mohan**, Chairman, Madras Diabetes Research Foundation (MDRF), Chennai. MD, PhD, FRCP; 600+ peer-reviewed publications on Indian diabetes & nutrition.
>
> *Paraphrased, not a direct quote. Live pipeline verifies exact wording against published work before publish.*

— MEMORY ANCHOR —

"Magnesium is your nervous system's volume knob. Turn it too low, and everything feels louder than it is."

Sources: Abbasi et al. 2012, *J Res Med Sci* · de Baaij et al. 2015, *Physiol Rev* · DiNicolantonio et al. 2018, *Open Heart* · Zhang et al. 2016, *Hypertension* · Cochrane Database · ICMR Dietary Guidelines for Indians · research publications of MDRF / V. Mohan.